



CHITTAGONG UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Biomedical Engineering

Automatic Medicine Reminder Box using Arduino

Course Title: Digital Logic Design Sessional

Course Code: EEE 228

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Project title:

Digital Medicine Reminder Box using Arduino.

Objectives:

1. Detail the design and implementation of an Arduino-based medicine reminder box.
2. Explain the user interface development for ease of interaction.
3. Describe the scheduling system for flexible medication plans.
4. Discuss the utilization of wireless connectivity for remote monitoring and accessibility.

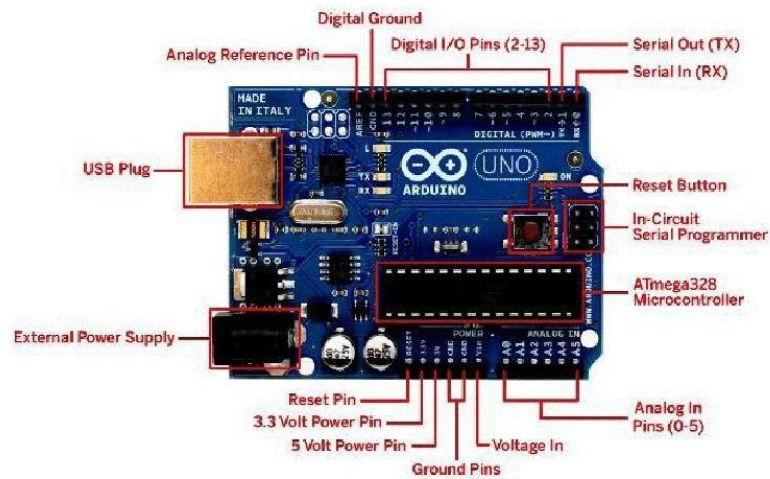
Required components:

Components Required for Automatic Medicine reminder using Arduino

1. Arduino Uno (We can use other Arduino boards also, like Pro mini, Nano)
2. RTC DS3231 module
3. 16x2 LCD Display
4. Buzzer
5. Led(any color)
6. Breadboard
7. Push Buttons
8. 10K Potentiometer
9. 10K,1K Resistors
10. Jumper Wires

Description Of Components:

1.Arduino Uno (We can use other Arduino boards also, like Pro mini, Nano)



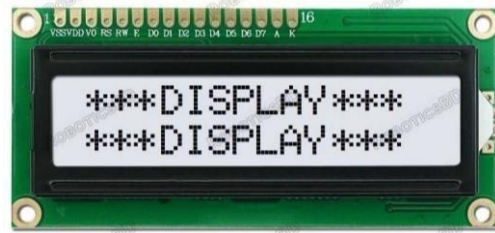
-It is a microcontroller development board. which are used here as the brain of the controller of this project.

1. RTC DS3231 module



-RTC3231 is a real-time clock system designed by a chip name rtc3231 it is having a coin battery that continuously powers the module due to this continuous power it remembers the time once you set the time.

3. 16x2 LCD Display



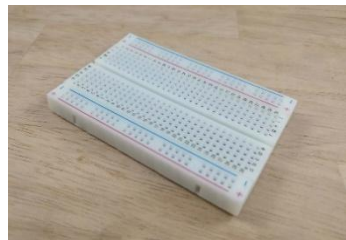
-16×2 displays are common uses in the project there are 16 columns and 2 rows that's why it is known as 16×2 LCD.

4. Buzzer



-It makes noise whenever we power it. basically, it converts electrical energy into sound energy

6. Breadboard

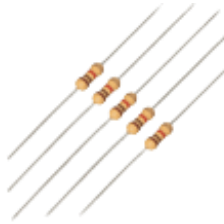


-to implement the circuit

7. 10k potentiometer



-The 10k potentiometer adjusts parameters, possibly display brightness or buzzer volume, enhancing user control in the medicine reminder box project. 8 .10k,1k Resistors



- To control current flow.

9. Jumper Wires(male to male & male to females)



-To construct the circuit.

Block Diagram:

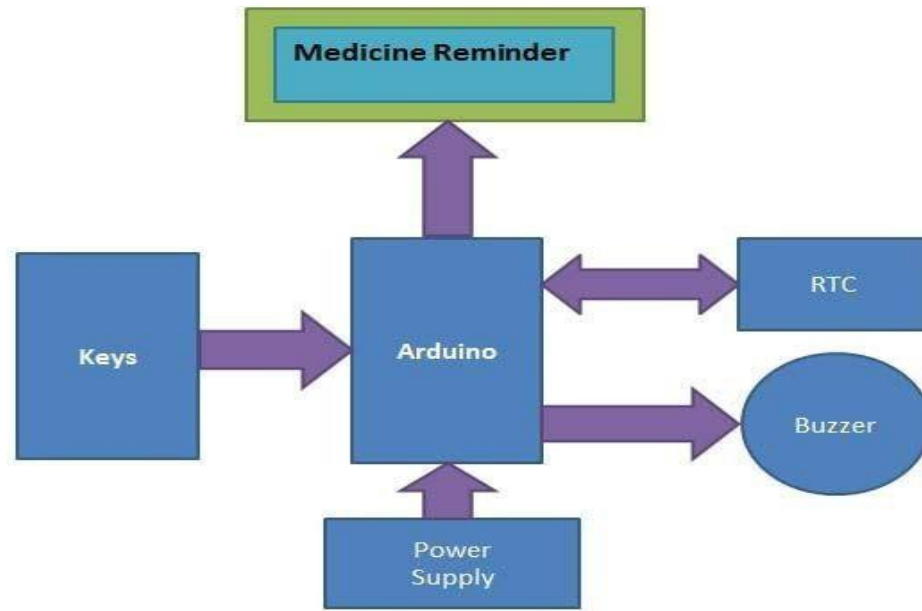


Fig 01 : Block Diagram of Digital Medicine Reminder Box

Circuit Diagram:

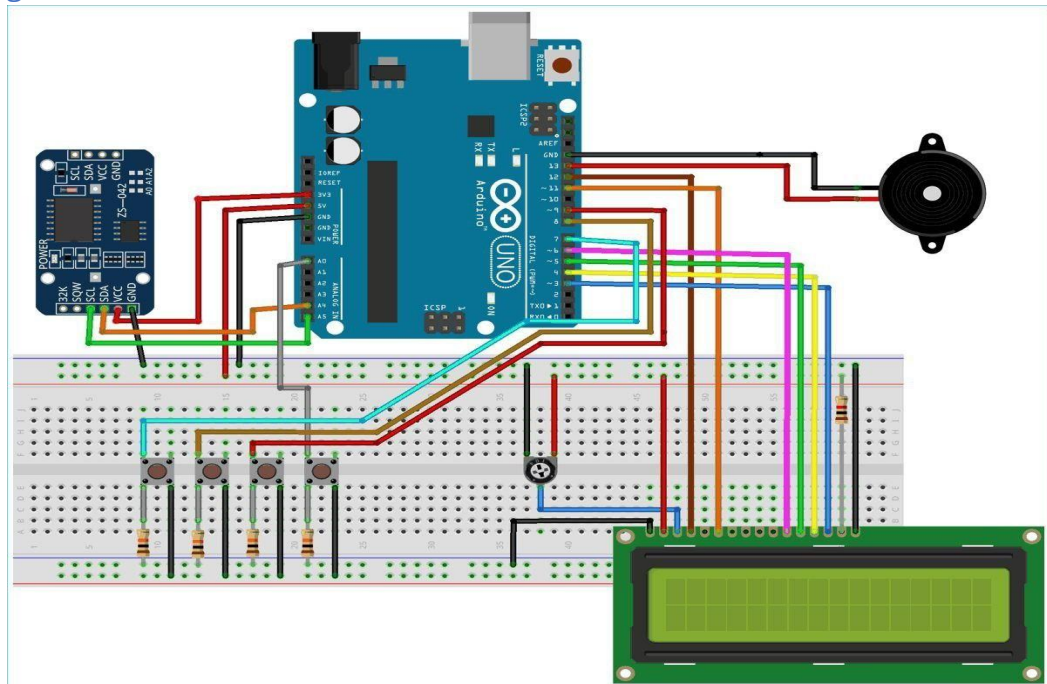


FIG 02 : Circuit Representation

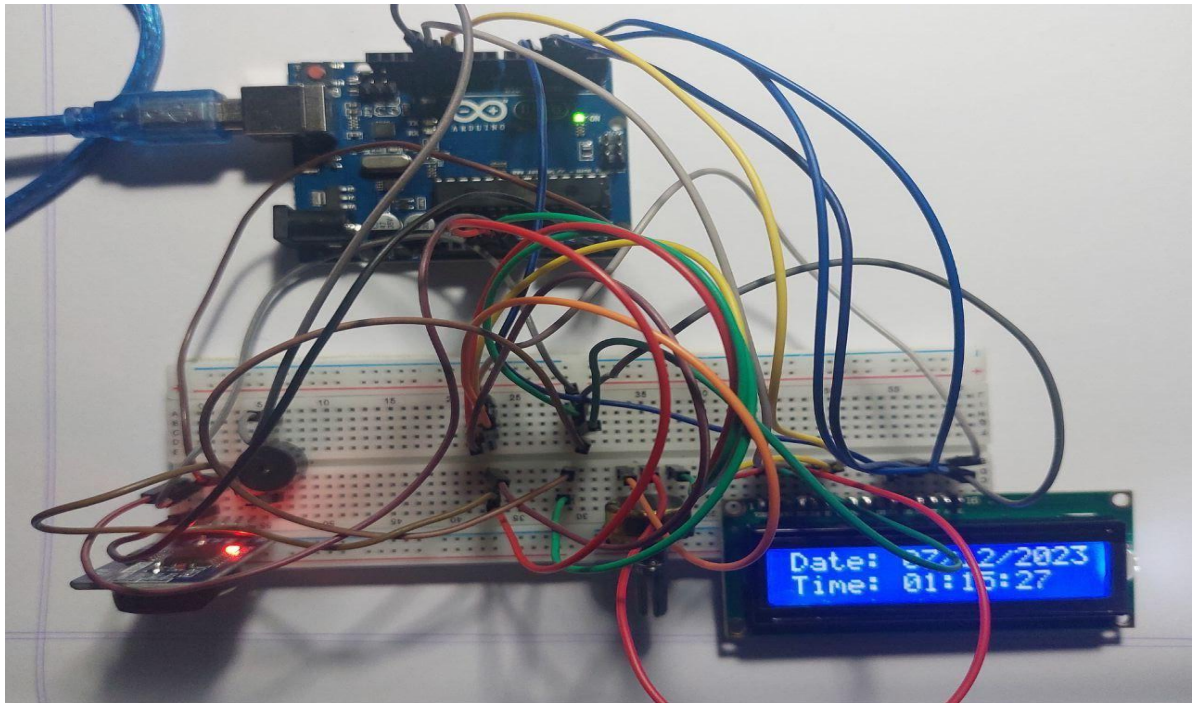


FIG 03: Regular Display

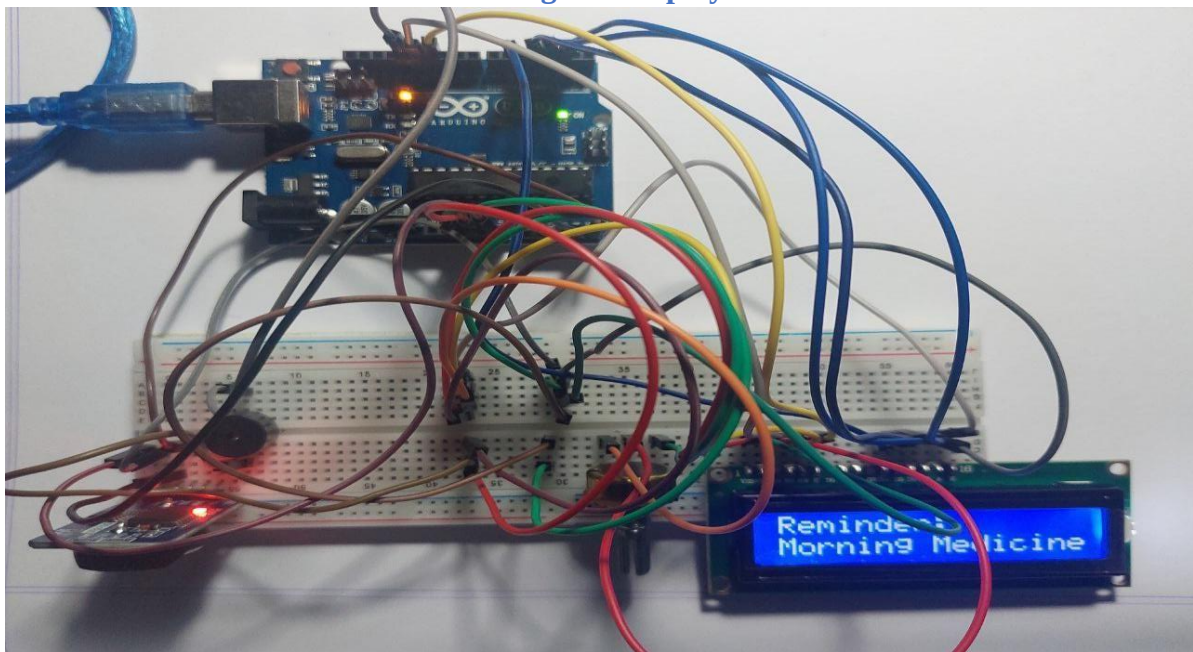


FIG 04: During Reminder

Procedure:

1. Connect Arduino pins to peripheral [ins as follows:

- Pin 2 to D7 of 16x2 LCD Display
- Pin 3 to D6 of 16x2 LCD Display
- Pin 4 to D5 of 16x2 LCD Display
- Pin 5 to D4 of 16x2 LCD Display
- Pin 11 to EN pin of 16x2 LCD Display
- Pin 12 to RS pin of 16x2 LCD Display
- Pin 13 to +Ve Pin of Buzzer and LED
- A4 to SDA of DS3231
- A5 to SCL of DS3231
- 3.3V to Vcc of DS3231
- Gnd to Gnd

2. Upload the following code into the Arduino Uno

Source code:

```
#include <Wire.h>
#include <RTClib.h>
#include <LiquidCrystal.h>

RTC_DS3231 rtc;

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int buzzerPin = 13; // Change this to the pin where you connected the buzzer

void setup() {
  Serial.begin(9600);
```

```
    if (!rtc.begin()) {  
        Serial.println("Couldn't find RTC");  
    while (1);  
    }  
  
    // Uncomment and set the following line with the correct date and time  
    rtc.adjust(DateTime(2023, 12, 7, 1, 15, 23));  
  
    lcd.begin(16, 2);  
  
    // Display welcome message  
    lcd.print("Digital Medicine");  
    lcd.setCursor(0, 1);  lcd.print("  
BOX ");  
  
    delay(3000); // Display welcome message for 3 seconds  
  
    pinMode(buzzerPin, OUTPUT);  
}  
  
void loop() {  
    DateTime now = rtc.now();  
  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Date: ");  
    printTwoDigits(now.day());  
    lcd.print('/');
```

```
    printTwoDigits(now.month());  
    lcd.print('/');  
    lcd.print(now.year(), DEC);
```

```
    lcd.setCursor(0, 1);  
    lcd.print("Time: ");  
    printTwoDigits(now.hour());  
    lcd.print(':');  
    printTwoDigits(now.minute());  
    lcd.print(':');  
    printTwoDigits(now.second());
```

```
    checkReminders(now);
```

```
    delay(1000); // Update every second  
}
```

```
void checkReminders(DateTime now) {  if  
(now.hour() == 1 && now.minute() == 16) {  
    displayReminder("Morning Medicine");  
    activateBuzzer();  
    } else if (now.hour() == 18 && now.minute() == 0) {  
displayReminder("Evening Medicine");    activateBuzzer();  
    } else if (now.hour() == 22 && now.minute() == 0) {  
displayReminder("Goodnight Medicine");    activateBuzzer();  
    } else {  
    digitalWrite(buzzerPin, LOW); // Turn off the buzzer if no reminder  
    }  
}
```

```

void displayReminder(const char *reminder) {
  lcd.clear(); lcd.setCursor(0, 0);
  lcd.print("Reminder:"); lcd.setCursor(0, 1);
  lcd.print(reminder);
}

```

```

void activateBuzzer() {
  digitalWrite(buzzerPin, HIGH); // Turn on the buzzer
  delay(30000); // Buzzer will buzz for 1 minute (60000 milliseconds)
  digitalWrite(buzzerPin, LOW); // Turn off the buzzer
}

```

```

void printTwoDigits(int number) {
  if (number < 10) {
    lcd.print("0");
  }
  lcd.print(number, DEC);
}

```

Working of the Medicine Reminder:

In the medication box system, Arduino autonomously manages medication schedules and reminders without the need for push buttons. Users input timing and reminder details directly into the Arduino code, which coordinates with the DS3231 real-time clock. At programmed intervals, the system activates alerts through the buzzer and LED, ensuring timely medication reminders. This code-driven approach simplifies user interaction, offering a seamless and automated solution for effective medication management.